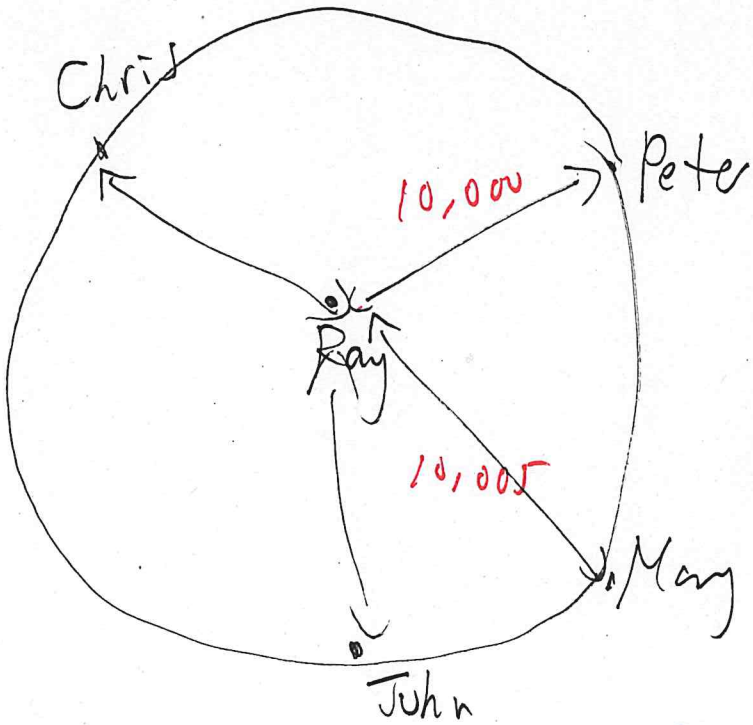
COMP 5331 (30 Sept)

	Computer	Chinese	..	.	.
Ray	100	70			
Peter	95	5			
Mary	40	71			

large

$dist(Ray, Peter) = \sqrt{(100-95)^2 + (70-5)^2 + \cdots}$ = large (= 10,000)

$dist(Ray, Mary) = \sqrt{(100-40)^2 + (70-71)^2 + \cdots}$ = large (= 10,005)



if X (itemset of size k) is freq.

X' (any proper subset of X) is freq.

proper itemset of X with size k-1

Ex.4 a)

	X	Y	Z
(11, 13, 5)	X2	Y2	Z1
(12, 11, 21)	X2	Y2	Z3
(11, 17, 27)	X2	Y2	Z3
(13, 14, 38)	X2	Y2	Z4
(22, 37, 36)	X3	Y4	Z4
(24, 31, 27)	X3	Y4	Z3
(25, 35, 21)	X3	Y4	Z3
(29, 34, 4)	X3	Y4	Z1
(35, 5, 4)	X4	Y1	Z1
(36, 6, 5)	X4	Y1	Z1

[1, 10]	X1	Y1	Z1
[11, 20]	X2	Y2	Z2
[21, 30]	X3	Y3	Z3
[31, 40]	X4	Y4	Z4

threshold = 40%

For Size 1,

For attribute X,

unit	density
X1	0
X2	0.4
X3	0.4
X4	0.2

For attribute Y

unit	density
Y1	0.2
Y2	0.4
Y3	0
Y4	0.4

For attribute Z

unit	density
Z1	0.4
Z2	0
Z3	0.4
Z4	0.2

dense unit →

L_1: dense unit = {(X2), (X3)}

dense unit = [(Y2), (Y4)]

dense unit = {(Z1), (Z3)}

sub-space → L_1 = {[X], [Y], [Z]}

2- dimensional dense unit generation

Candidate Generation

Join / Prune step

subspace → $C_2 = \{ [X,Y], [X,Z], [Y,Z] \}$

For [Y,Y]

dense unit → $C_2 = \{ (X2,Y2)^{0.4}, (X2,Y4)^{0}, (X3,Y2)^{0}, (X3,Y4)^{0.4} \}$

For (X, Z)

$C_2 = \{ (X2,Z1)^{0.1}, (X2,Z3)^{0.2}, (X3,Z1)^{0.1}, (X3,Z3)^{0.2} \}$

For [Y, Z],

$C_2 = \{ (Y2,Z1)^{0.1}, (Y2,Z3)^{0.2}, (Y4,Z1)^{0.1}, (Y4,Z3)^{0.2} \}$

Dense Unit Generation

Count step

dense unit → $L_2 = \{ (X2,Y2), (X3,Y4) \}$

$L_2 = ($ $)$

$L_2 = ($ $)$

sub-space → $L_2 = ([X,Y])$

3- dimensional dense unit generation

Candidate Generation

Join step

subspace → $C_3 = ($ $)$

subspace containing dense units

$= \{ [X], [Y], [Z], [X,Y] \}$

(b) For attr. X

dense unit = ((X2), (X3))

cluster 1 : X = [11, 30]

For attr Y,

dense unit = ((Y2), (Y4))

cluster 1 : Y = [11, 20]

cluster 2 : Y = [31, 40]

For attr. Z,

dense unit = ((Z1), (Z3))

cluster 1 : Z = [1, 10]

cluster 2 : Z = [21, 30]

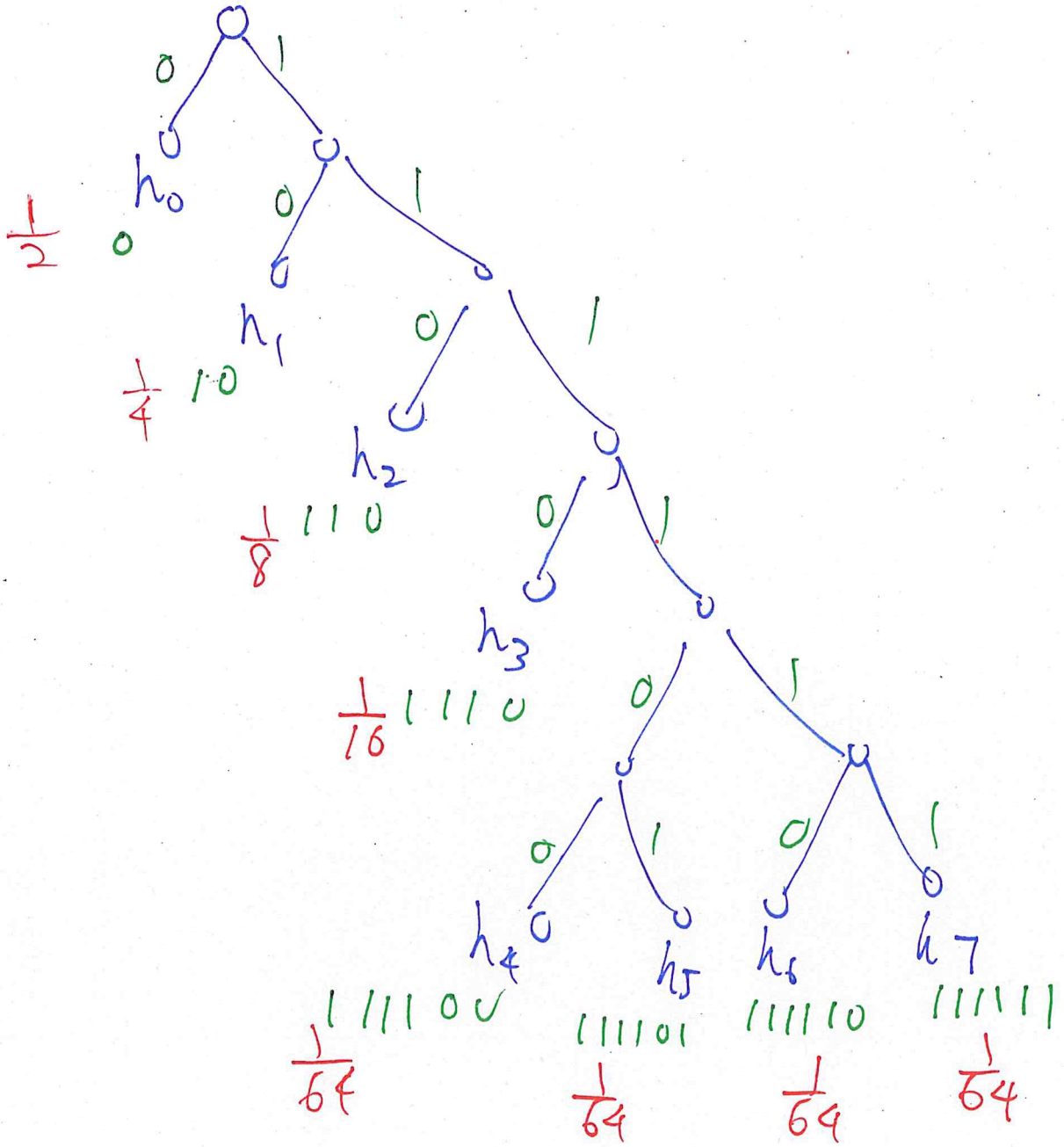
For attr (X, Y),

dense unit = [(X2, Y2), (X3, Y4)]

cluster 1 { X = [11, 20]
Y = [11, 20]

cluster 2 { X = [21, 30]
Y = [31, 40]

		← bit		
0	0	0	h_0	$\frac{1}{2}$
0	0	1	h_1	$\frac{1}{4}$
0	1	0	h_2	$\frac{1}{8}$
0	1	1	h_3	$1/16$
1	0	0	h_4	$1/64$
1	0	1	h_5	$1/64$
1	1	0	h_6	$1/64$
1	1	1	h_7	$1/64$


0
1
h_0
0
$\frac{1}{2}$
0
1
h_1
0
1
$\frac{1}{4}$
10
h_2
$\frac{1}{8}$
110
0
1
h_3
$\frac{1}{16}$
1110
0
1
0
1
0
1
h_4
h_5
h_6
h_7
111110
$\frac{1}{64}$
111101
$\frac{1}{64}$
111110
$\frac{1}{64}$
111111
$\frac{1}{64}$

expected length of string

$= \frac{1}{2} \times 1 + \frac{1}{4} \times 2 + \frac{1}{8} \times 3$ $+ \frac{1}{16} \times 4 + \left(\frac{1}{64} \times 6\right) \times 4$

$= 2$ bits

$-\log_2 \frac{1}{2}$, $-\log_2 \frac{1}{4}$, $-\log_2 \frac{1}{8}$, $-\log_2 \frac{1}{16}$, $-\log_2 \frac{1}{64}$